covalent bond. For example, in $K_4[Fe(CN)_6]$, the central metal ion is Fe^{2+}. In the coordination entity $[Fe(CN)_6]^{4-}$, the Fe^{2+} accepts an electron pair from each ligand, CN^- and thereby forming six coordinate covalent bonds with them. since, the central metal ion has an ability to accept electron pairs, it is referred to as a Lewis acid.

5.3.3 Ligands:

The ligands are the atoms or groups of atoms bound to the central atom/ion. The atom in a ligand that is bound directly to the central metal atom is known as a donor atom. For example, in $K_4[Fe(CN)_6]$, the ligand is CN^- ion, but the donor atom is carbon and in $[Co(NH_3)_6]Cl_3$ the ligand is NH_3 molecule and the donor atom is nitrogen.

Coordination sphere:

The complex ion of the coordination compound containing the central metal atom/ion and the ligands attached to it, is collectively called coordination sphere and are usually enclosed in square brackets with the net charge. The other ionisable ions, are written outside the bracket are called counter ions. For example, the coordination compound $K_4[Fe(CN)_6]$ contains the complex ion $[Fe(CN)_6]^{4-}$ and is referred as the coordination sphere. The other associated ion K^+ is called the counter ion.

Coordination polyhedron:

The three dimensional spacial arrangement of ligand atoms/ions that are directly attached to the central atom is known as the coordination polyhedron (or polygon). For example, in $K_4[Fe(CN)_6]$, the coordination polyhedra is octrahedral. The coordination polyhedra of $[Ni(CO)_4]$ is tetrahedral.

Coordination number:

The number of ligand donor atoms bonded to a central metal ion in a complex is called the coordination number of the metal. In other words, the coordination number is equal to the number of σ-bonds between ligands and the central atom.

For example,

i. In $K_4[Fe(CN)_6]$, the coordination number of Fe^{2+} is 6.

ii. In $[Ni(en)_3]Cl_2$, the coordination number of Ni^{2+} is also 6. Here the ligand 'en' represents ethane-1,2-diamine (NH_2-CH_2-CH_2-NH_2) and it contains two donor atoms (Nitrogen) Each ligand forms two coordination bonds with nickel. So,totally there are six coordination bonds between them.

Oxidation state (number):

The oxidation state of a central atom in a coordination entity is defined as the charge it would bear if all the ligands were removed along with the electron pairs that were shared with the central atom. In naming a complex, it is represented by a Roman numeral. For example, in the coordination entity $[Fe(CN)_6]^{4-}$, the oxidation state of iron is represented as (II). The net charge on the complex ion is equal to the sum of

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